

Placement Empowerment Program
Cloud Computing and DevOps Centre

Set Up a Cloud-Based Monitoring Service

Enable basic cloud monitoring (e.g., CloudWatch on AWS). View metrics like CPU usage and disk I/O for your cloud VM.

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Introduction and Overview

Cloud-based monitoring is essential for tracking system performance, ensuring uptime, and proactively identifying potential issues. AWS CloudWatch is a comprehensive monitoring service that allows users to collect and analyze logs, track system metrics, set alarms, and automate responses to performance changes.

This PoC explores the process of enabling CloudWatch monitoring for an EC2 instance, viewing key system metrics such as CPU utilization and disk I/O, and setting up alarms for proactive management. The objective is to enhance visibility into system performance and ensure that resources are optimized efficiently.

Objective

- To enable AWS CloudWatch monitoring for an EC2 instance and ensure that system performance data is available for real-time analysis.
- To track critical performance metrics such as CPU utilization, memory usage, disk I/O, and network activity, helping administrators monitor the health of the instance.
- To configure alarms that notify administrators when predefined thresholds are breached, allowing for quick intervention to prevent system slowdowns or failures.
- To assess CloudWatch's capabilities in improving cloud resource management by analyzing logs and creating actionable insights.

Importance

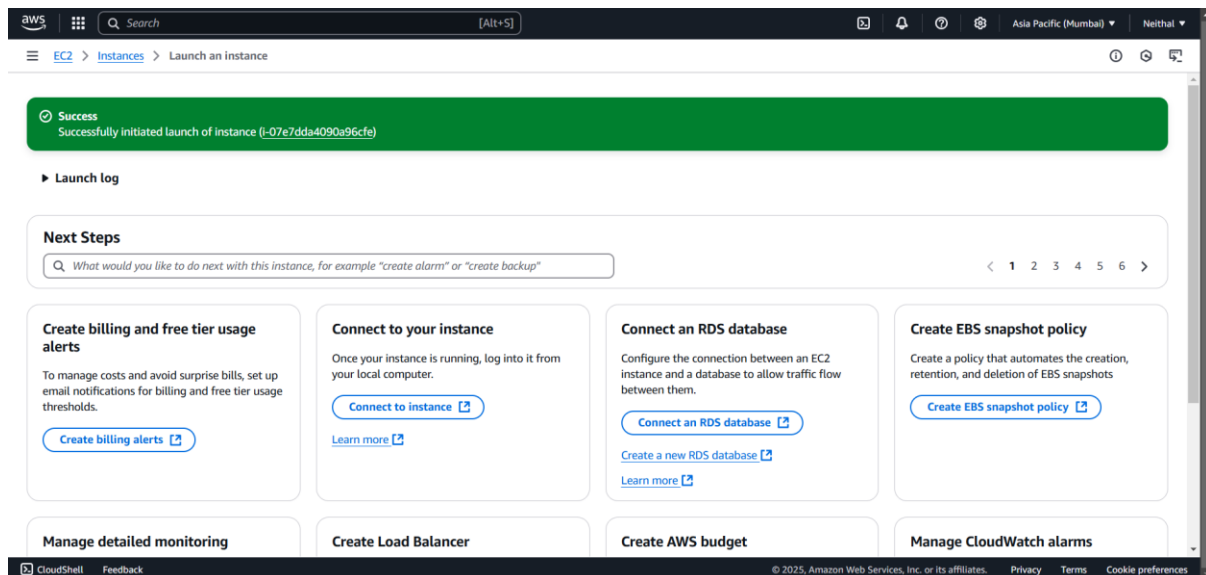
Monitoring cloud resources is crucial for ensuring optimal performance, maintaining system health, and responding to issues before they impact users. AWS CloudWatch enables businesses and cloud administrators to gain real-time insights into system performance, set up automated alarms to detect anomalies such as high CPU usage or excessive memory consumption, and trigger automated actions or notifications. Storing logs for analysis helps in understanding system behavior over time and identifying patterns that could lead to failures. Effective monitoring can significantly reduce downtime and

operational costs by optimizing cloud resource usage based on real-time and historical performance data. Additionally, CloudWatch improves security by identifying unusual patterns in system activity, such as unexpected spikes in network traffic.

Steps for Execution

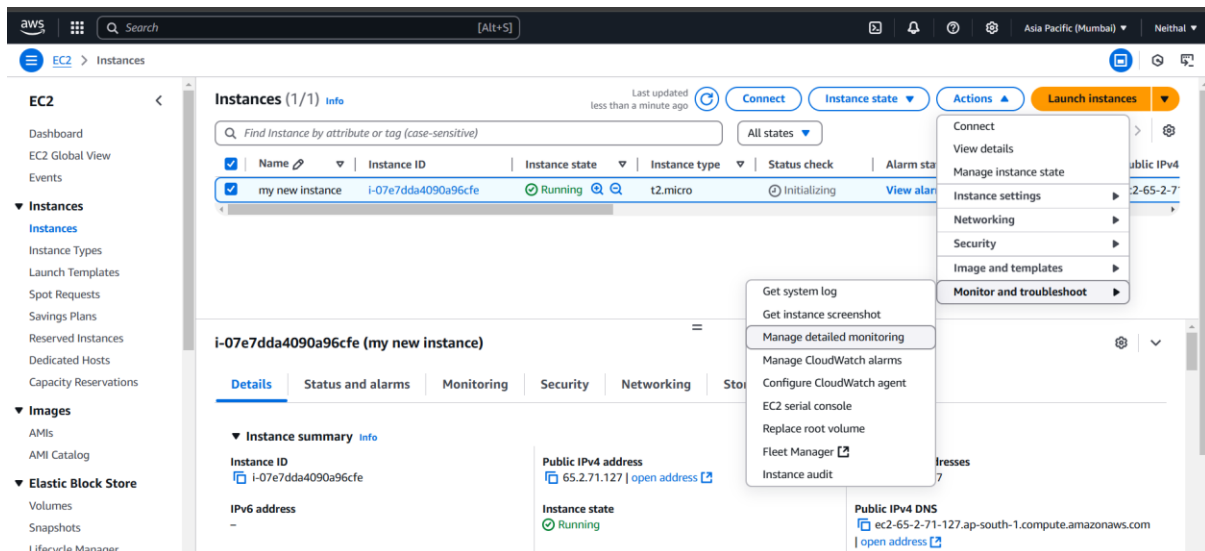
Step 1:

Access the AWS Management Console and navigate to the EC2 Dashboard. Select the EC2 instance that requires monitoring and enable detailed monitoring to collect data at one-minute intervals instead of the default five-minute intervals. This ensures more accurate performance tracking. Once detailed monitoring is activated, CloudWatch will start collecting key performance metrics.



Step 2:

Open the CloudWatch service from the AWS Console and navigate to the Metrics section. Under EC2 Metrics, select the instance ID to view important performance indicators such as CPU utilization, disk read/write operations, and network traffic. Reviewing these metrics over different time periods can help identify trends and detect performance bottlenecks.



Step 3:

Set up a CloudWatch Alarm for high CPU usage. Within CloudWatch, navigate to the Alarms section and create an alarm based on EC2 per-instance metrics. Select CPU Utilization as the metric and define a threshold, such as 80% usage. Configure the alarm to send notifications via Amazon SNS or trigger automated actions like stopping or restarting the instance. Once created, the alarm will continuously monitor CPU usage and alert administrators when the threshold is exceeded.

Detailed monitoring

After you enable detailed monitoring for an instance, monitoring data is available in 1-minute periods. [Learn more](#)

Instance ID

 **i-07e7dda4090a96cfe** (my new instance)

Detailed monitoring

☒ Enable

 After you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period for the instance.

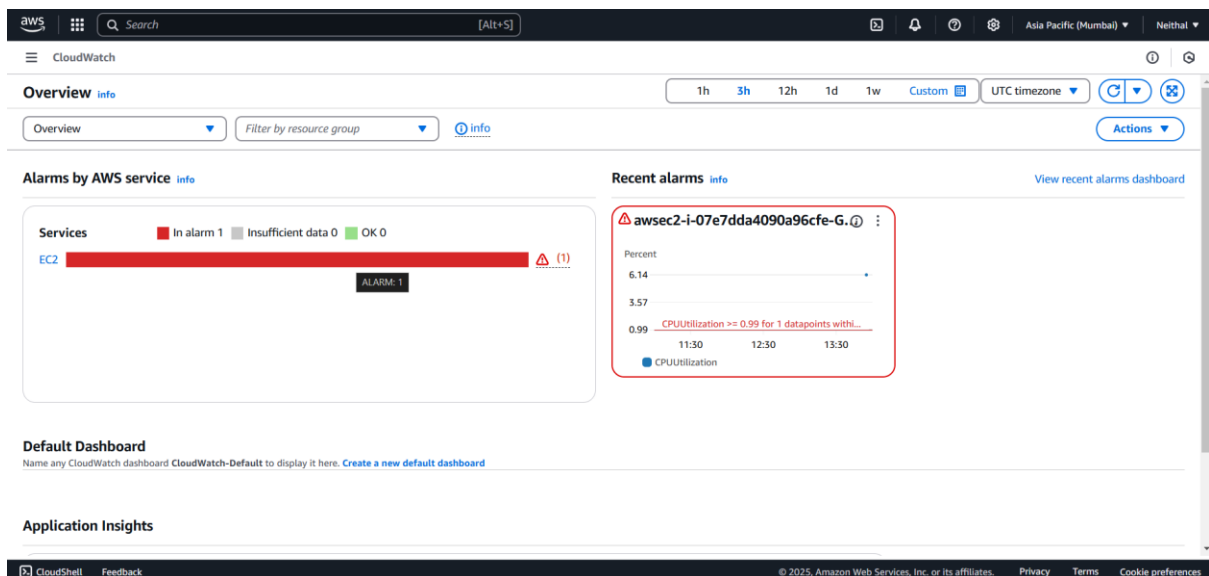
[Additional charges apply](#)

Cancel

Confirm

Step 4:

Conduct a test by running CPU-intensive processes on the EC2 instance to verify the effectiveness of the alarm. If CPU usage surpasses the defined threshold, an alert should be triggered, notifying the specified recipients. Based on the received notifications, administrators can fine-tune the alarm settings to prevent unnecessary alerts while ensuring critical issues are flagged.



Step 5:

Regularly review logs and metrics to optimize monitoring configurations. Utilizing CloudWatch Dashboards can provide centralized visualization of multiple instances and services, making it easier to track system health. Additionally, enabling Log Insights allows for deeper analysis of system behavior. For further automation, integrate AWS Lambda to execute predefined actions when alarms are triggered, such as automatically scaling resources to manage increased workloads.

Conclusion

This PoC successfully demonstrates the setup and usage of AWS CloudWatch for monitoring EC2 instances. By enabling CloudWatch monitoring, administrators can gain real-time insights into system performance, set up proactive alerts, and optimize cloud resource utilization. CloudWatch enhances operational efficiency and minimizes downtime risks through automated monitoring and alerts. Implementing a well-configured monitoring strategy ensures that instances remain stable, performant, and cost-efficient.

What I Learned

- AWS CloudWatch provides a robust solution for real-time monitoring of cloud resources, enabling administrators to track system health efficiently.
- Setting up alarms ensures that potential performance issues are detected early, preventing major system failures and downtime.
- Detailed monitoring allows for more granular data collection, improving analysis accuracy and helping administrators make informed scaling decisions.
- Proper alert configurations help in proactive cloud resource management by notifying stakeholders about performance issues before they impact operations.
- CloudWatch dashboards offer centralized visibility for multiple AWS resources, improving cloud infrastructure monitoring and making it easier to track performance trends over time.
- Automating incident responses with AWS Lambda can improve system resilience and reduce the need for manual intervention during performance issues.

By leveraging AWS CloudWatch effectively, businesses can improve cloud resource management, enhance security, and optimize infrastructure performance while minimizing costs.